

Junction Branch Decisions and Geometry Violations

Droplet rollout model comparison

July 14, 2026

1 Purpose

This note documents the junction branch-decision diagnostic and the follow-up geometry-violation analysis. The goal is to separate two related but distinct failure modes:

- **Decision error:** the model commits to the wrong outgoing branch at the junction.
- **Physical geometry error:** the predicted droplet footprint leaves the admissible channel mask, measured by ellipse outside-channel fraction.

The analysis uses saved validation rollout predictions from:

`outputs/models/rollout_model_comparison/rollout_predictions.npz`

No model inference or retraining is involved.

2 Junction Event Definition

True junction events are identified from the true future trajectory only. A trajectory is counted as a true junction event when it has at least one valid assignment to an explicitly allowed incoming branch before committing to an outgoing branch. The outgoing branch commitment rule is:

same outgoing branch assignment for 3 consecutive evaluable rollout steps.

The outgoing branches are `left` and `right`. The incoming branch used in this validation set is `inlet`. Arbitrary non-outgoing branches are not treated as incoming, and a trajectory that begins already assigned to `left` or `right` is not counted as a true junction event.

The corrected event finder identified 4,990 true junction events:

- `inlet -> left`: 1,555 events
- `inlet -> right`: 3,435 events

3 Branch Decision Results

4 Branch Confusion Matrices

The confusion matrices below include only committed decisions: `correct` and `wrong_branch`. `no_commitment` and `unclassifiable` events are excluded from these matrices.

Table 1: Junction decision classification by model.

Model	Events	Correct	Wrong	No commit	Wrong rate	Noncommit rate
History-20	4,990	4,715	243	32	4.9%	0.6%
Markovian	4,990	4,514	387	89	7.9%	1.8%
Geometry-aware	4,990	4,498	435	57	8.8%	1.1%

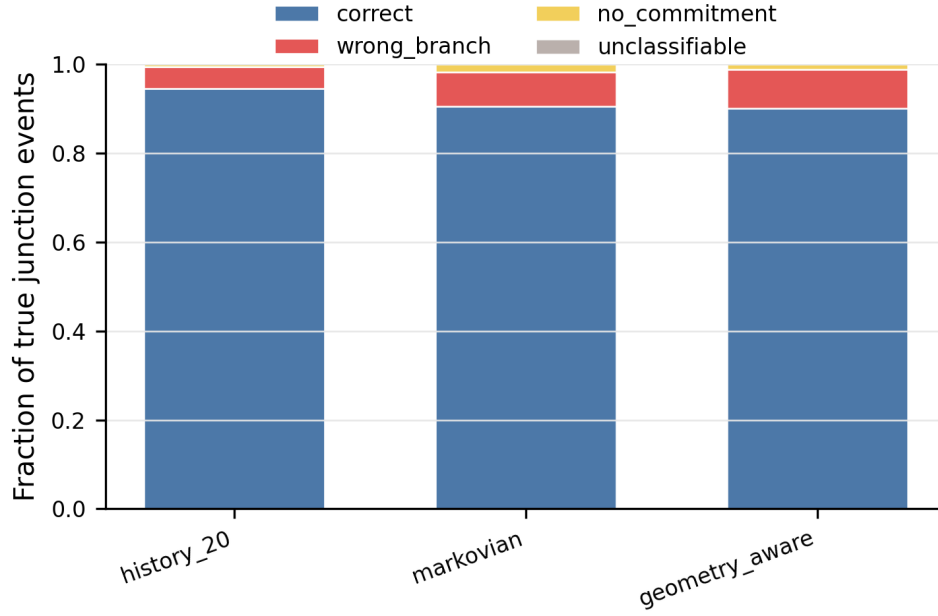


Figure 1: Decision outcome composition for true junction events. Each bar is normalized by the number of true junction events for that model.

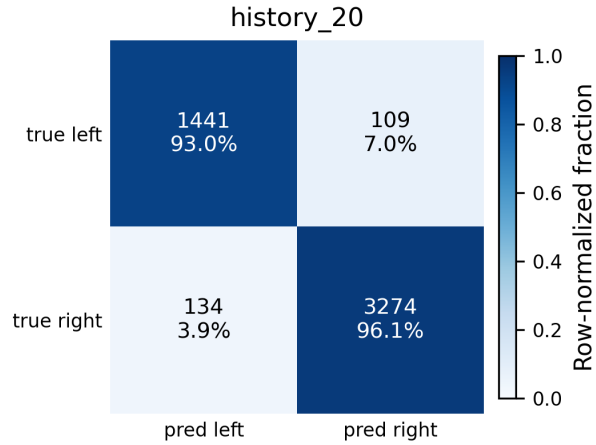


Figure 2: History-20 branch-decision confusion matrix.

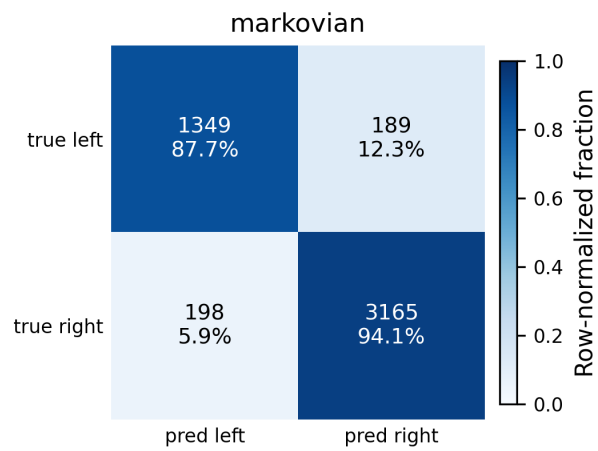


Figure 3: Markovian branch-decision confusion matrix.

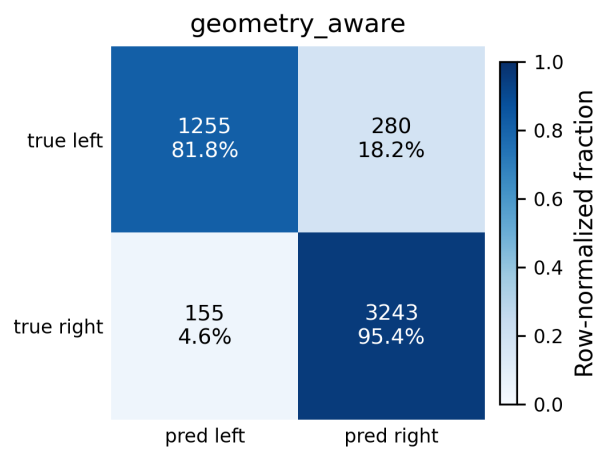


Figure 4: Geometry-aware Markovian branch-decision confusion matrix.

5 Geometry Violations for Wrong Branch Decisions

A geometry violation is defined as a predicted droplet ellipse outside-channel fraction greater than the configured geometry tolerance. The table below asks: among droplets that committed to the wrong outgoing branch, how many also violated the channel geometry?

Table 2: Geometry violations among wrong-branch decisions.

Model	Wrong branch	Violating	Fraction	Median peak outside
History-20	243	142	58.4%	0.030
Markovian	387	244	63.0%	0.053
Geometry-aware	435	225	51.7%	0.021

The geometry-aware model has more wrong-branch decisions in this saved run, but a smaller fraction of those wrong decisions leave the channel. Its median peak outside-channel fraction is also lower than the Markovian baseline. This suggests that the geometry-aware model may reduce physically impossible wrong rollouts even when it does not correct the branch-choice decision itself.

6 Geometry Violations for Correct Branch Decisions

The same geometry test was repeated for droplets whose predicted branch decision was correct. This asks whether the model chooses the right outgoing channel but still places the droplet footprint outside the admissible channel.

Table 3: Geometry violations among correct-branch decisions.

Model	Correct branch	Violating	Fraction	Median peak outside
History-20	4,715	687	14.6%	0.000
Markovian	4,514	1,596	35.4%	0.005
Geometry-aware	4,498	2,801	62.3%	0.046

This is the surprising result. In this saved prediction set, the geometry-aware model often predicts the correct outgoing branch but still violates the channel footprint criterion. This points to a placement or wall-clearance failure rather than a branch-selection failure: the model may be choosing the correct channel while placing droplets too close to, or partially outside, the detected channel boundary.

7 Output Files

Primary junction-decision outputs:

```
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_metrics.csv
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_events.csv
outputs/models/rollout_model_comparison_junction_smoke/junction_decision_outcomes.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_history_20.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_markovian.png
outputs/models/rollout_model_comparison_junction_smoke/junction_confusion_geometry_aware.png
```

Geometry follow-up outputs:

```
outputs/models/rollout_model_comparison_wrong_decision_geometry/  
outputs/models/rollout_model_comparison_correct_decision_geometry/
```

8 Interpretation

The decision and geometry diagnostics together show that branch choice and physical admissibility are not the same error mode. The history-based model has the best branch-decision accuracy and the lowest rate of geometry violation among correct decisions. The Markovian model is worse on both branch decisions and physical placement. The geometry-aware model reduces the severity of wrong-branch geometry violations, but in this saved run it has a high rate of geometry violations even for correct branch decisions.

The next useful diagnostic is to inspect geometry-aware correct-branch events with large outside-channel fractions. These cases should reveal whether the model is systematically hugging one wall, cutting corners at the junction, or using bounding-box footprints that are too large for the local channel width.